



MEK/TKD

Laboratory 1

Introduction to Mechatronics, MECH4201

A/Prof Antonio Padilha Lanari Bo

OSLO METROPOLITAN UNIVERSITY
STORBYUNIVERSITETET



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Summary

This document is a complete manual for the Laboratory 1 in the Introduction to Mechatronics (MECH4201) course at OsloMet. This lab will involve a range of materials covered in the first part of the course, in particular electronic instrumentation and signal conditioning applied to sensing systems.

In this manual, you will find a general description of the activities to be performed and what you are expected to achieve, as well as a report template and marking criteria. Minor amendments may be made to these tasks at any time, so please use only the last version of this document¹. Also, additional resources, including necessary concepts and methods, are found within the lecture material.

¹ Check the Version Control table at the end of the document.

1 Guidelines

1.1 Safety precautions and instructions

- Be calm and relaxed while working in the lab.
- Keep the work area neat and clean.
- Never use any faulty or damaged equipment and tools.
- Avoid using long wires, since they may get in your way while changing your circuit.
- Ideally, do not wear bracelets, rings, and metal watch bands near an energized circuit. Be mindful of them otherwise.
- When working with energized circuits, use only one hand while keeping the rest of your body away from conducting surfaces.
- Always connect elements in your circuit from load to power supply. I.e. the power supply should be connected last.
- Always check your circuit connections before power it ON.
- After the session, return all equipment to their original storage location.
- If an individual comes in contact with a live electrical conductor:
 - o Do not touch the equipment, the cord, the person.
 - o Turn off or disconnect the power supply and pull out the conductor using insulated material.
- No food or drink in the lab.
- Be aware of the locations of fire extinguishers and first aid kit.

1.2 General description

This laboratory activity (from now on simply referred to as Lab 1) involves the design of an electronic system to condition signals from a resistive sensor. By the end of Lab 1, you will:

1. Have an overview of basic principles and common practices related to the design, implementation and testing of electronic systems.
2. Understand two essential circuits in electronic instrumentation: the Wheatstone bridge and the instrumentation amplifier.
3. Build and test a signal conditioning system for strain gauges.
4. Compare the performance of the developed system against a commercial alternative.

Lab 1 comprises of three different activities: (i) the design and analysis of an electronic circuit, including specification of components values, (ii) the implementation and testing of this circuit in a laboratory setting, and (iii) the preparation of a report, which includes answering additional questions. Experimental activities should be completed in pairs, while (i) and (iii) should be completed individually.

While this section in the manual describes all the Lab 1 activities, the next section presents a report template and the marking criteria for this assessment.

1.3 Equipment and experimental protocol

The following equipment and components will be used:

- Breadboard and passive components (wires, cables, connectors, etc)
- Resistors (4x 120 Ω , 6x 10 k Ω , 1x 100 k Ω , possibly other values)
- 1x TL074 operational amplifier
- Strain gauge and related equipment
- Power supply, function generator, oscilloscope, multimeter

1.4 Pre-lab design

1.4.1 Wheatstone bridge

Question D1

Strain gauges are elements whose resistance R_s change in response to strain. These elements are adhesively bonded onto surfaces, enabling the measurement of strain of the material of interest. The following model describes the changes in electrical resistance as a function of strain:

$$\varepsilon = \frac{dR}{RG'} \quad (1)$$

where ε represents strain, R is the nominal resistance, dR is the change in resistance, and G is known as the gauge factor.

The strain gauge used in the final test have a nominal resistance of 120 Ω and $G = 2$. Calculate the change in resistance if a strain of 1% is applied.

Question D2

To convert the change in resistance of the strain gauge into a voltage output V_{out} , we use a Wheatstone bridge circuit (Figure 1). The strain gauge resistance is represented by R_s ,

$$R_s = R + dR. \quad (2)$$

The three other resistors in this circuit have the same resistance as the nominal resistance of the strain gauge, i.e. R .

The bridge output is approximately proportional to the change in resistance dR . Derive the equation that provides V_{out} as a function of dR , R and V_b .

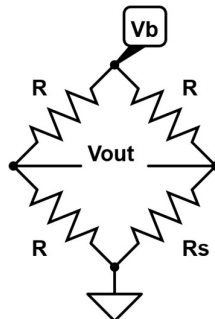


Figure 1. Wheatstone bridge

Question D3

Unfortunately, the resistors in our lab have a tolerance of $\pm 5\%$, which will produce an offset voltage V_{offset} at the output of the bridge. In addition, the relative change in resistance that we want to observe is very small, often on the order of 0.1% . This means that if we arbitrarily choose some $\pm 5\%$ resistors, we will have some difference V_{offset} already before we apply strain to obtain the voltage change due to the variable resistance dR .

Consider now a bridge excitation voltage V_b of 5 V and the nominal resistance defined in D1 (i.e. $120\ \Omega$). Calculate the output voltage for a strain of 1% . Re-calculate V_{out} considering now a specific scenario where the three bridge resistors feature a resistance of $114\ \Omega$.

1.4.2 Instrumentation amplifier

Question D4

As observed in D3, the resulting V_{out} from the Wheatstone bridge is often very small. Hence, voltage amplification is needed for proper processing. Here, we will use an instrumentation amplifier, which is illustrated in Figure 2.

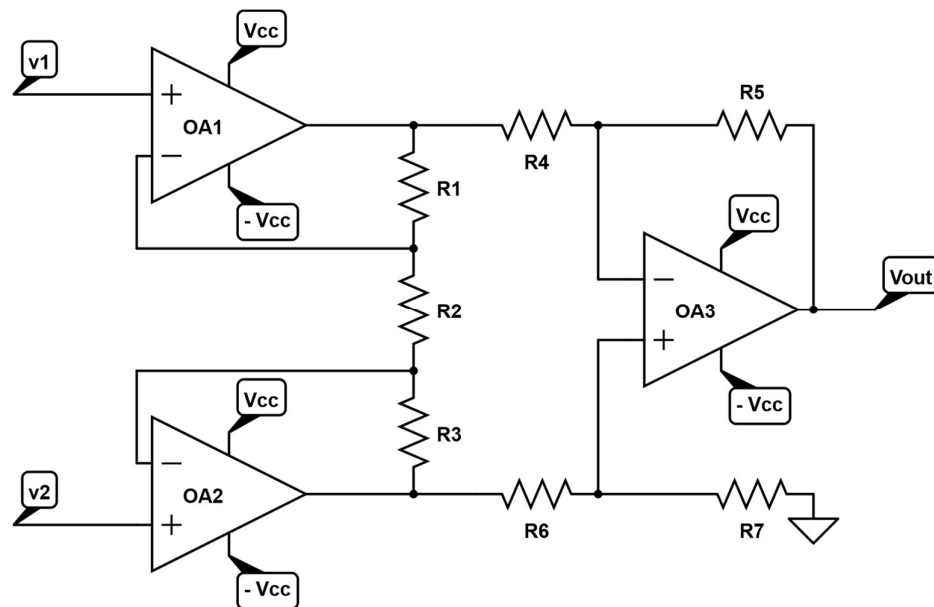


Figure 2. Instrumentation amplifier

Derive the transfer function of an instrumentation amplifier.

Question D5

Now consider the operational amplifiers (opamps) that compose an instrumentation amplifier are powered using $V_{\text{cc}} = \pm 12\text{ V}$. Given a perfectly balanced bridge and that the strain gauge resistance may vary up to 0.5% , propose a gain of the instrumentation amplifier circuit that will provide $V_{\text{out}} = \pm 9\text{ V}$. Considering $R1 = R3 = R4 = R5 = R6 = R7 = 10\text{ k}\Omega$, what will be the resistance of $R2$ to obtain this gain?

1.5 Laboratory

1.5.1 Wheatstone bridge

Task L1

Firstly, build the Wheatstone bridge according to your design. Initially, use resistances R of same nominal value of R_s . Use $V_b = 12\text{ V}$. Once the circuit is built, measure the output voltage V_{out} and report the value.

Considering the imperfections on R and R_s , the output voltage V_{out} may be different from 0 V . If the measured V_{out} deviates by more than 5 mV , measure the voltage across all resistances and add them to your report. Then, pick a new resistor R and replace one of the original resistors.

An alternative to reduce this problem is to add a variable resistor (e.g., a potentiometer) to the circuit. This can be done by replacing R connected to the ground by a potentiometer. Using now $V_b = 5\text{ V}$, build this circuit and adjust the potentiometer to obtain $V_{out} = 0\text{ V}$.

Before proceeding to the next circuit, show your results to the instructor.

1.5.2 Instrumentation amplifier

Task L2

Once the bridge is built, it is necessary to amplify the output voltage. The circuit, illustrated in Figure 2, uses three opamps within a TL074 integrated circuit. Use $V_{cc} = 12\text{ V}$, $R_1 = R_3 = R_4 = R_5 = R_6 = R_7 = 10\text{ k}\Omega$. In your implementation, however, start by building the output stage, illustrated in Figure 3. Build the circuit sufficiently apart from your Wheatstone bridge, where a function generator is used to test the amplifier.

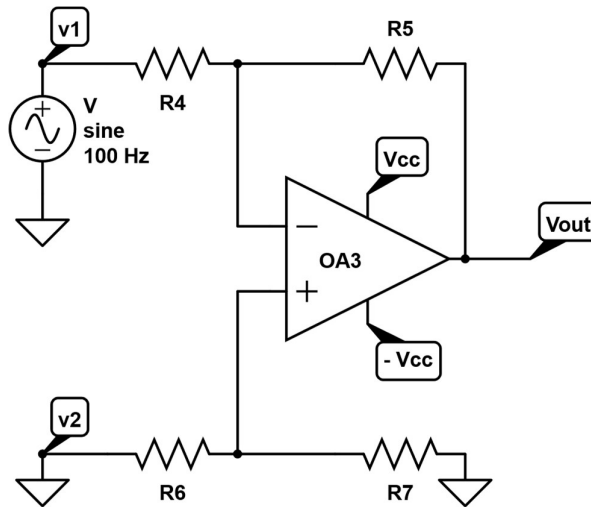


Figure 3. Difference amplifier and sine wave as input

Before proceeding to the next circuit, show your results to the instructor.

Task L3

Next, proceed to build the rest of the instrumentation amplifier, as depicted in Figure 4. A function generator is used to test the circuit before connection to the bridge. Record the input and output signals and add to the report.

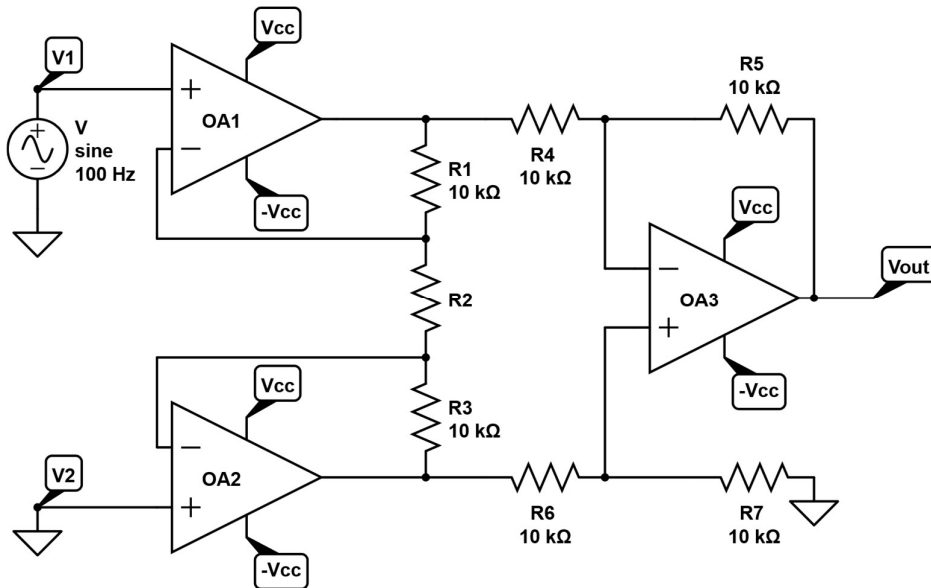


Figure 4. Instrumentation amplifier and sine wave as input

Show your results to the instructor.

1.5.3 Scale calibration

Task L4

Once both the Wheatstone bridge and the instrumentation amplifier have been validated, it is time to test the system using the strain gauge setup. With instructor assistance, connect the strain gauge to your circuit and visualize the output voltage V_a . It should be close to 0 V when no load is applied.

Afterwards, move on to the scale calibration procedure. Wait for the measurement to stabilize before taking note of it. Create a calibration table by applying loads of known weights and recording the corresponding voltage. Report the result.

The same procedure will be performed using the native acquisition system at the lab. Also take notes of the readings and create a calibration table.

In cases issues are observed in this stage, a system built by the instructor may be used.

Task L5

In order to test your sensing system for new loads, loads of unknown weight will be applied by the instructor. Record and report the measured voltages V_u .

The same procedure will be performed using the native acquisition system at the lab. Also take notes of the reading.

1.6 Post-lab questions

1.6.1 Wheatstone bridge

Question P1

List at least one desirable and one undesirable effect an increase in the magnitude of V_b (i.e. the bridge excitation voltage) have on the measurements made with a Wheatstone bridge?

Question P2

Explain another potential issue when measuring resistance using Wheatstone bridges and propose a method to minimize its effect.

1.6.2 Instrumentation amplifier

Question P3

Cite and explain one reason why instrumentation amplifiers may not feature zero common-mode rejection ratio (CMRR).

Question P4

Search for ICs that already possess all the functionalities of an instrumentation amplifier encapsulated. Discuss potential advantages and disadvantages of using this IC or general opamps, as used in this lab.

1.6.3 Scale calibration

Question P4

Calculate two calibration curves using the data collected in the lab for the two measurement systems. Use curve fitting tools. Graphically illustrate the collected data and the fitted curve. What are the scale sensitivities?

Question P5

Using the calibration curve calculated in P4, determine the unknown weights using both recorded voltages V_u obtained in L5.

2 Report template & Marking criteria

2.1 Pre-lab design (20 marks)

2.1.1 Wheatstone bridge

Question D1 (2 marks)

Question D2 (4 marks)

Question D3 (4 marks)

2.1.2 Instrumentation amplifier

Question D4 (3 marks)

Question D5 (4 marks)

2.2 Laboratory (55 marks)

2.2.1 Wheatstone bridge

Task L1 (10 marks)

2.2.2 Instrumentation amplifier

Task L2 (20 marks)

Task L3 (20 marks)

2.2.3 Scale calibration

Task L4 (4 marks)

Task L5 (1 marks)

2.3 Post-lab questions (25 marks)

2.3.1 Wheatstone bridge

Question P1 (2 marks)

Question P2 (3 marks)

2.3.2 Instrumentation amplifier

Question P3 (3 marks)

Question P4 (2 marks)

2.3.3 Scale calibration

Question P5 (10 marks)

Question P6 (5 marks)

2.4 Additional references

References

Alciatore, David. *Introduction to Mechatronics and Measurement Systems*. McGraw-Hill, 5th edition, 2018

Bentley, John. *Principles of Measurement Systems*. McGraw-Hill, Pearson Education, 4th edition, 2005

Version control

File	Version	Date	Author	Comments
mech4201_lab1_manual	1	02.09.2024	A/Prof Antonio Padilha L Bo	First version
	2	09.09.2024	A/Prof Antonio Padilha L Bo	Changes in Sec. 1.5, including one new figure
	3	10.09.2024	A/Prof Antonio Padilha L Bo	Corrections to Sec. 1.5.2
	4	13.09.2024	A/Prof Antonio Padilha L Bo	Changes in Secs. 1.5, 1.6, and 2
	5	22.08.2025	A/Prof Antonio Padilha L Bo	Updates in Sec. 1.5 and marks distribution in Sec. 2.2
mech4201_lab1_reporttemplate	1	02.09.2024	A/Prof Antonio Padilha L Bo	First version
	2	13.09.2024	A/Prof Antonio Padilha L Bo	Changes in Sec. 1.5